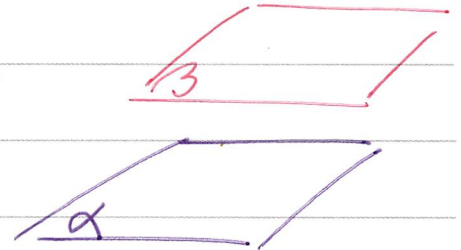
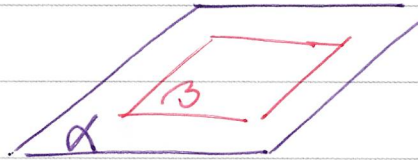
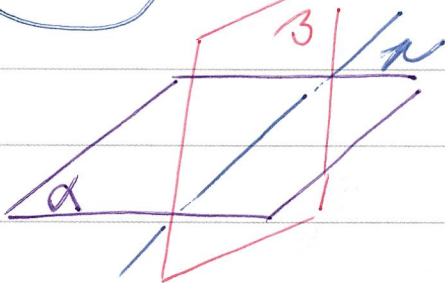


ALGORITMY

FR9

VZÁJOMNÁ POLOHA 2 ROVÍN



ROZNOBEŽNÉ

$$\vec{n}_\alpha \nparallel \vec{n}_\beta$$

TOTOŽNÉ

$$\vec{n}_\alpha \parallel \vec{n}_\beta$$

ROVNOBEŽNÉ

$$\vec{n}_\alpha \parallel \vec{n}_\beta$$

PRETÍNajú SA
V PRIAMKE

∞

$\emptyset$

$$\alpha: -5x + 3y - 2z - 1 = 0$$

$$\beta: 2x + 3y + z - 4 = 0$$

ROZNOBEŽNÉ $\Rightarrow$

$\vec{n}_\alpha (-5, 3, -2)$; $\vec{n}_\beta (2, 3, 1) \Rightarrow$ PRETÍNajú
SA V PRIAMKE

$$-5x + 3y - 2z - 1 = 0 \quad (-1)$$

$$2x + 3y + z - 4 = 0 \quad \downarrow$$

$$7x + 3z - 3 = 0$$

$\Rightarrow$ VOĽÍM napr. $z = t$

$$\Rightarrow 7x = 3 - 3t$$

$$x = \frac{3}{7} - \frac{3}{7}t$$

dosadíme za x a z napr. do α



ALGORITMY

$$-5 \left(\overset{x}{\frac{3}{7} - \frac{3}{7}t} \right) + 3y - \overset{z}{2t} - 1 = 0$$

$$-\frac{15}{7} + \frac{15}{7}t + 3y - 2t - 1 = 0$$

$$3y = \frac{15}{7} - \frac{15}{7}t + 2t + 1$$

$$3y = \frac{22}{7} - \frac{1}{7}t \quad / : 3$$

$$y = \frac{22}{21} - \frac{1}{21}t$$

$$p: x = \frac{3}{7} - \frac{3}{7}t$$

$$y = \frac{22}{21} - \frac{1}{21}t$$

$$z = t; t \in \mathbb{R}$$

JEDNODUCHÁ KONTROLA:

ZVOĽME SI NA PR. $t=1$ A DOSAÐME DO ROVNICE

$$\text{PRIAMICY } p \Rightarrow x=0$$

$$\left. \begin{aligned} y &= \frac{22}{21} - \frac{1}{21} = \frac{21}{21} = 1 \\ z &= 1 \end{aligned} \right\} P[0,1,1]$$

$$\alpha: 0 + 3 - 2 - 1 = 0 \checkmark; \beta: 0 + 3 + 1 - 4 = 0 \checkmark \quad \text{DOSAÐME DO } \alpha, \beta$$

BOD p PATRÍ DO PRIENIKU OBOCH ROVÍN ✓